Quantitative Methods for Business and Economics

Lecture 1 - Introduction and foundational concepts

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Introduction: Mathematics for Business and Economics

- Mathematics per se **not** a distinct field in business and economics
- We use mathematics **as an approach** to study reality
- It is not hard to think about business and economic models which were initially not mathematical
  - Consumer choice through utility maximization
  - Firm pricing and profit maximization
- Modern researchers represent models using math
  - Allows to extend the model using mathematical "laws"
  - Makes models less subject to interpretation
  - Allows for statistical testing of hypotheses

Is it so challenging?

- Math is often feared because of its complexity
- Often the reason people fail is poor foundations
- Must master basic concepts before moving to more complex ones
  - This means that math takes time and few have the patience for it
  - Comparing yourself to others is **not useful**
  - Some have better foundations, some have inherent talent, some have more time

Order of operations

- The order of operations defines the priority of operations in an expression
- This is a consensus which everyone agreed to so we can all understand each other
    - Calculators and statistical software follow this order
    - As basic as this is, this is tested on the GRE (graduate school entrance exam)
- **PEMDAS**: **P**arentheses, **E**xponents, **M**ultiplication and **D**ivision, **A**ddition and **S**ubtraction
- If two operations have the same priority, we go from left to right.
- No sign means multiplication

Exercise

Order of operations

Calculate the following expression:

$$3 + 4 \times 2^2(3 + 1 \times 3 \div 3) + 4 - 5 + 5$$

Solution

Order of operations

- Start with the parentheses. Apply PEMDAS within too.

$$4 \times 1 = 4$$

- Substitute into the full expression, follow order, and solve

$$\begin{aligned} 3 + 4 \times 2^2(4) + 4 - 5 + 5 \\ 3 + 4 \times 4(4) + 4 - 5 + 5 \\ 3 + 4 \times 16 + 4 - 5 + 5 \\ 3 + 64 + 4 - 5 + 5 \\ 67 + 4 - 5 + 5 \\ 71 - 5 + 5 \\ 71 \end{aligned}$$

Thoughts

- As useless as this might be with numbers, it is crucial when working with variables
  - E.g. What does $\lambda_1(x_1 + x_2) \times \lambda_1 x_1 + x_2^2(\tau \cdot \gamma)$ simplify to?
- Calculators typically don't work with variables
- Not understanding PEMDAS may lead you to code the wrong formula in your software
  - It is hard to debug this as it is not a syntax error

Variables and the greek alphabet

- Variables are used to represent unknowns in equations
  - Technically, they are placeholders for numbers
- x is commonly used, y and z are common too
- Scientists need more variables as they study more complex systems
  - The greek alphabet is used for this
- Get comfortable seeing and using the greek alphabet

The Greek alphabet

Alpha - Α α	Eta - Η η	Nu - Ν ν	Tau - Τ τ
Beta - Β β	Theta - Θ θ	Xi - Ξ ξ	Upsilon - Υ υ
Gamma - Γ γ	Iota - Ι ι	Omicron - Ο ο	Phi - Φ φ
Delta - Δ δ	Kappa - Κ κ	Pi - Π π	Chi - Χ χ
Epsilon - Ε ε	Lambda - Λ λ	Rho - Ρ ρ	Psi - Ψ ψ
Zeta - Ζ ζ	Mu - Μ μ	Sigma - Σ σ/ς	Omega - Ω ω

Figure 1: Greek alphabet. Source: University of Northern Colorado

- Commonly, the same letter is used for different purposes in different fields and contexts
  - β represents the slope coefficient, but also the discount factor in finance

Working with percentages and proportions

- Percentages are a way to express a fraction of a whole (the symbol being %)
- The word “percent” means “per hundred”; meaningful for calculation

$$\text{Percentage}\% = \frac{\text{Part}}{\text{Whole}} \times 100\%$$

- A useful way of thinking about percentages is as *proportions*: the decimal representation of the percentage
  - Divide by 100 and drop the % sign
  - 50% is 0.50, 25% is 0.25, 100% is 1.00
- This is useful for calculations so you don’t use the above formula all the time
- Example: A 25% discount on a $75 item is $0.25 \times 75 = 18.75$

Working with percentages and proportions

- Example: What is the final price of a \$75 item with a 25% discount?
- You may be tempted to use the rule of three, but this is not necessary (it takes longer)
- Use proportions to understand that 1 represents the full price (the \$75) and 0.25 represents the discount.

$$\$75 \cdot (1 - 0.25) = \$75 \cdot 0.75 = \$56.25$$

- Notice that the new final price is 75% of the original price, so this makes sense!
    - Don't be afraid to use your intuition
    - Business analysts and economists are not math machines, we use intuition to guide us

Percentage change and subindices

- It is common to express increases and decreases in percentages
- This involves introducing the concept of **variation**
  - Typically, we express change or variation as Δ
- Often, we use subindices to represent the old and new values
  - E.g. x_0 is the old value, x_1 is the new value
- The formula for percentage change is

$$\Delta = \frac{\text{New} - \text{Old}}{\text{Old}} = \frac{x_1 - x_0}{x_0}$$

- This gives a **proportion** (e.g. 0.10 for a 10% raise). Multiply by 100 to get the familiar percentage figure (e.g. 10%)
- A shorthand for this is dividing the final value by the initial value and subtracting 1

Solving for a final value after a percentage change

- Example: A company gives employees a 10% raise. If an employee currently earns \$46,000 per year, what would the new salary be?
- Typically, one would calculate 10% of \$46,000 and add it to the original value.

$$x_0 + (x_0 \cdot \Delta) = x_1$$

- It is faster to add Δ to 1 and multiply by the original value $(1 + 0.10 = 1.10)$
- Doing this is a simplification from the formula above
- However, it is also intuitive, since you're simply finding a value which is 10% larger than the whole (which is 1).
- So, the new salary is \$46,000 $\times$ 1.10 $=$ \$50,600

Solving for a final value after a percentage change

- The same can be done for percentage *decreases*
- Example: A store's weekly revenue has decreased by 5%. If revenue was \$500, what is the new weekly revenue?
- The new weekly revenue is \$500 $\times$ (1-0.05) $=$ \$500 $\times$ 0.95 $=$ \$475
- Once again, this is intuitive: the new revenue is only 95% of the original revenue

An exercise for you

- Example: A coffee drink in Toronto costs \$3.50 (CAD) after a 15% sales tax. How much tax is the consumer paying?

Solution

- The stated price (\$3.50) is the total paid after tax, so the pre-tax price P_0 satisfies $P_0 \cdot 1.15 = 3.50$
- Solving: $P_0 = 3.50/1.15 \approx 3.04$, so the tax is $3.50 - 3.04 \approx \$0.46$
- If instead you computed $15\% \times \$3.50 = \$0.525$ — that is wrong, because the tax applies to P_0, not the final price
- Key lesson: always be clear about what the "whole" is in the percentage formula

Ratios

- Ratios are a way to express the relationship between two parts of a whole
  - Not the same as the percentage, which is the relationship of a part to the whole
- They are often expressed as 1 to 2, 1:2.
- These can be a bit confusing, but context is helpful.
  - A ratio of 1:2 means that for every 1 unit of the first part, there are 2 units of the second part
- Example: A company allocates budget between R&D and marketing in a ratio of 1:2, meaning for every $1 in R&D, $2 goes to marketing.

Ratios

- You can work out the relationship from one part to the whole by adding the parts of the ratio.
  - This may or may not make sense depending on the context
- With a ratio of 1:2 for R&D and marketing, the whole is 3 parts, but this is not meaningful on its own
- However, if you're told that in a hospital the ratio of doctors to nurses is 1:2, then doctors are $1/3$ of the whole staff

Application: market share and percentage change interpretations

- Suppose Company A's market share rises from 12% to 15% in a quarter
- Question for you: which of the following statements is correct?
  1. Market share increased by 3 percentage points
  2. Market share increased by 25%
  3. Market share increased by 15%

Percentage points and percent changes

- When the quantity being analyzed is itself a percentage, we need to distinguish two concepts
- **Percentage point change**: the absolute difference — $15\% - 12\% = 3$ percentage points (statement 1)
- **Percent change**: the relative change — $(15 - 12)/12 \times 100 = 25\%$ (statement 2)
- Both statements 1 and 2 are correct; they measure the same change in different ways
- Statement 3 is wrong — 15% is the new market share, not the change

Equations

- An equation is a statement that two expressions are equal, typically with an unknown.
  - E.g. $x + 2 = 5$
- We often care about solving equations, which means finding the value of the unknown that makes the equation true (i.e. satisfies the equation)
  - For above, $x = 3$ will make the equation true ($5 = 5$)
- Equations can be solved by isolating the unknown on one side of the equation, and doing the math.

Equations

- There are rules for solving equations:
    - You can add or subtract the same value to both sides
    - You can multiply or divide both sides by the same value
    - Exponents must be raised to the same power on both sides
    - You can take the square root of both sides by adding $\pm$
    - We will see more rules as we go along

Functions

- A function is a rule that assigns a unique output to each input
    - Think of it as a machine that takes an input and gives an output
    - We stop caring about solving for the unknown, and care about the relationship between the input and output
- Functions are often represented as $f(x)$, where f is the function and x is the input
    - The value of the function at x is $f(x)$
- A function doesn't necessarily have to be a mathematical formula, but it is often represented as one
- Economists and business analysts use functions to represent real-world relationships
    - E.g. the relationship between advertising and sales
    - E.g. the relationship between price and quantity demanded
- Functions can be linear, quadratic, exponential, logarithmic, etc.

Functions

- The simplest function is the constant function, which always returns the same value
  - E.g. $f(x) = 5$
- Notice how this challenges our equation solving skills
  - There is no unknown to solve for
  - If we input 3, we get 5, if we input 10, we get 5
- A linear function is the next simplest.
  - E.g. $f(x) = 2x + 3$
  - Here we do have an input x which goes into the function
  - The function multiplies the input by 2, adds 3, and returns the result
  - It's called linear because the graph of the function is a straight line

Functions: domain and range

- The **domain** of a function is the set of all possible inputs
  - For $f(x) = 2x + 3$, the domain is all real numbers ($\mathbb{R}$)
- The **range** of a function is the set of all possible outputs
  - For $f(x) = 2x + 3$, the range is all real numbers ($\mathbb{R}$)
- The domain and range are important to understand the behavior of the function

Functions: domain and range

- Some functions have a restricted domain because of the way the equation is defined
  - E.g. $f(x) = \frac{1}{x}$ has a domain of all real numbers except 0
  - The function is not defined at 0 because there is no such thing as division by 0
- The domain and range are often represented as intervals
  - E.g. $x \in (-\infty, 0) \cup (0, \infty)$
  - E.g. $f(x) \in (-\infty, 0) \cup (0, \infty)$
- Without differential calculus, to find the domain and range of a function easily, you can plot it
  - With some time, you will start recognizing the behavior of functions

Inequalities

- Inequalities are statements that two expressions are not equal
  - E.g. $x > 5$
- When we have more elaborate expressions, we can have more complex inequalities
  - E.g. $2x + 3 > 5$
  - Simplify using the rules of equations to find which values of x make the inequality true.
  - E.g. $2x > 2$, $x > 1$. This means that all values of x greater than 1 make $2x + 3 > 5$ true.
- Use the same rules as equations to solve inequalities, except when multiplying or dividing by a negative number
  - In this case, the inequality sign flips
  - E.g. $-2x < 6$, $x > -3$
- Inequalities are often used to represent constraints in optimization problems